

SW Engineering CSC 648/848

Projectory

Portfolio and AI Career Guidance Platform

Section 06 — Team 03

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Milestone 1

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1. Executive Summary

College students often have useful skills and projects but still have trouble connecting what they have done in school to a possible career direction. Their work is usually spread across resumes, class projects, personal websites, and other platforms, while career advice is often separated from the work that actually demonstrates their abilities. Projectory is intended to bring these pieces together in one place so that students can both present their work and better understand where it may lead them.

Projectory is a portfolio and AI career guidance platform designed for college students. Students will be able to build a portfolio containing their education, skills, experience, career interests, and projects. The platform will use information from the portfolio to provide personalized career guidance, including possible career paths, strengths, areas for improvement, and suggested next steps. Students will also be able to attach supported interactive 3D models to selected projects, giving them a way to present certain work visually instead of relying only on text, links, and images.

The main value of Projectory is the combination of these features. The portfolio shows what a student has accomplished, the 3D project showcase gives students a more interactive way to demonstrate selected work, and the AI guidance uses the student's own portfolio information to help identify possible directions for the future. Instead of treating a portfolio and career planning as separate activities, Projectory connects them so students can show what they have done and use that information to think about what they should do next.

Team 03 is a six-person student development team with defined roles across product direction, Scrum coordination, GitHub, AI, backend, and frontend development. The team is using the milestone process to refine Projectory iteratively. During the next stage, the team will focus on validating the portfolio data flow, testing the AI career guidance workflow, and creating an initial proof of concept for displaying an interactive 3D project in the browser.

2. Personas and User Stories

The following personas represent several types of college students who may use Projectory. They are based on the original persona work prepared by the team and have been consolidated to focus on the main ways students will use the planned product.

Persona 1 – The Undecided Student

Name: Alex Martinez

Age: 18

Education: First-year college student

Behavior: Alex attends college but has not selected a clear career direction. Alex explores different subjects and frequently searches online for information about careers.

Interests: Finding a career that matches personal interests and provides good long-term opportunities.

Skills: Basic computer skills and experience using websites and mobile applications.

Pain Points: Alex feels overwhelmed by the number of majors and careers available and does not know where to begin.

Goal with Projectory: Build a starting portfolio and use career guidance to explore possible directions based on interests, education, skills, and projects.

Persona 2 – The Student Who Chose a Major but Not a Career

Name: Maya Patel

Age: 21

Education: Junior college student

Behavior: Maya has selected a major but is unsure what jobs and career paths are available after graduation. She researches careers online and talks with classmates about possible options.

Interests: Understanding how her degree, skills, and projects can connect to different career paths.

Skills: Comfortable researching information online and using university resources.

Pain Points: Maya understands her major but does not have a clear career plan or know which parts of her background are strongest.

Goal with Projectory: Organize her background into a portfolio and receive guidance about possible career paths, strengths, and areas to improve.

Persona 3 – The Student Preparing to Graduate

Name: Daniel Kim

Age: 23

Education: College senior

Behavior: Daniel is approaching graduation and is preparing his projects, experience, and skills for employers and other professional opportunities.

Interests: Presenting his work clearly, understanding career readiness, and identifying useful next steps before graduation.

Skills: Academic knowledge in his field and some experience creating resumes and applying for jobs.

Pain Points: Daniel is unsure whether he is presenting his work effectively and whether his current skills and projects support the career direction he wants.

Goal with Projectory: Create a shareable portfolio, showcase selected projects including interactive 3D work when appropriate, and use AI guidance to identify strengths and improvements.

Persona 4 – The Career-Changing Student

Name: Chris Johnson

Age: 32

Education: Returning college student

Behavior: Chris previously worked in another industry and returned to college to prepare for a different career.

Interests: Finding a new career direction while making use of transferable skills and previous experience.

Skills: Professional work experience, communication skills, and transferable skills from a previous career.

Pain Points: Chris does not know which previous skills will transfer to a new career or what additional skills and projects would strengthen the transition.

Goal with Projectory: Combine previous experience with current education and projects in one portfolio and receive guidance about possible paths and next steps.

Initial User Stories

US1 – Portfolio Creation – High Priority

As a student, I want to create a portfolio containing my education, skills, experience, career interests, and projects so that I can present my academic and professional background in one place.

US2 – Project Showcase – High Priority

As a student, I want to add projects and supporting media to my portfolio so that I can demonstrate work I have completed.

US3 – Interactive 3D Project Showcase – Mid Priority

As a student, I want to attach an interactive 3D model to a supported project so that viewers can explore my work in a more visual way.

US4 – Portfolio-Based Career Guidance – High Priority

As a student, I want Projectory to analyze my portfolio so that I can understand possible career paths related to my education, skills, experience, and projects.

US5 – Strengths and Areas for Improvement – High Priority

As a student, I want to understand strengths and areas for improvement identified from my portfolio so that I know what I should continue developing.

US6 – Suggested Next Steps – High Priority

As a student, I want personalized suggested next steps so that I can make informed decisions about skills, projects, experience, and career exploration.

US7 – Portfolio Sharing – Mid Priority

As a student, I want to share my portfolio with others so that they can view my background, projects, and supported interactive 3D models.

3. Data Definitions

User: A college student who uses Projectory to create and manage a portfolio, showcase projects, receive career guidance, and interact with platform features.

Portfolio: The collection of information that represents a student's academic and professional background. It may include education, skills, experience, career interests, projects, and other accomplishments.

Project: A piece of work created or completed by a student that can be displayed in the student's portfolio. A project may include a title, description, technologies used, images, links, and an optional interactive 3D model.

Skill: An ability or area of knowledge associated with a student. Skills may come from education, projects, work experience, or other activities and may be used by the AI career guidance feature.

Education: A student's academic background, such as university, degree program, major, relevant coursework, and expected graduation information.

Experience: Professional, academic, volunteer, internship, or other practical experience that contributes to the student's portfolio.

Career Interest: A professional field, role, or area that a student is interested in exploring. Career interests may be considered with portfolio information when generating career guidance.

3D Model: An interactive three-dimensional representation associated with a student's project. It allows a portfolio viewer to explore selected work in a more visual and interactive way than text or images alone.

Portfolio Viewer: A person who accesses a student's shared portfolio to view the student's background, projects, and supported 3D project models.

Career Guidance: Personalized information generated from a student's portfolio. It may include possible career paths, strengths, areas for improvement, and suggested next steps.

Career Path: A possible professional direction that may be relevant to a student based on the information in the portfolio.

Strength: A skill, experience, project, or other characteristic identified as a positive area in the student's portfolio.

Area for Improvement: A skill, experience, or qualification that may need further development to better prepare the student for a career goal.

Suggested Next Step: An action recommended to the student based on career guidance, such as learning a skill, improving a project, gaining experience, or exploring a career path.

4. Initial Functional Requirements

The following requirements describe the main functions currently planned for Projectory. They focus on what the system should allow the user to do and are linked to the initial user stories above.

ID	Title	Description	Priority	User Story
R1	User Registration	Allow students to create an account using basic information such as name, email, and password.	High	US1
R2	User Login and Logout	Allow registered users to securely log in and log out of their accounts.	High	US1
R3	Personal Information Management	Allow students to create, view, and update personal and academic information used in their portfolio.	High	US1
R4	Education Management	Allow students to add, edit, and remove education information such as school, major, degree, and graduation date.	High	US1
R5	Skills Management	Allow students to add, edit, and remove skills from their portfolio.	High	US1
R6	Experience Management	Allow students to add, edit, and remove work, internship, volunteer, or other relevant experience.	High	US1
R7	Project Management	Allow students to create, edit, view, and remove projects from their portfolio.	High	US2

R8	Project Media	Allow students to attach supporting content such as images and links to projects.	Mid	US2
R9	Portfolio Display	Organize the student's education, skills, experience, career interests, and projects into a unified portfolio page.	High	US1, US2
R10	AI Portfolio Analysis	Analyze information contained in the student's portfolio and generate a summary of the student's background.	High	US4
R11	Career Path Recommendations	Use portfolio information to suggest possible career paths that may align with the student's background and interests.	High	US4
R12	Strength Identification	Identify and present potential strengths based on information provided in the student's portfolio.	Mid	US5
R13	Improvement Recommendations	Identify areas where the student could improve and provide suggested next steps.	High	US5, US6
R14	Career Analysis Refresh	Allow students to request updated AI career guidance after changing their portfolio.	Mid	US4, US5, US6
R15	3D Model Attachment	Allow students to attach a supported 3D model to a portfolio project.	Mid	US3
R16	Interactive 3D Model Viewing	Allow portfolio viewers to interact with supported 3D project models using basic actions such as rotating and zooming.	Mid	US3
R17	Portfolio Sharing	Allow students to share their portfolio with others through a public or shareable portfolio page.	Mid	US7

5. Non-Functional Requirements

NFR1 – Usability: The platform should have a simple and intuitive interface that allows students to create and update portfolios without extensive technical knowledge. Navigation and important actions should be easy to find.

NFR2 – Performance: Most pages and portfolio information should load within approximately 2–3 seconds under normal conditions. AI guidance may take longer because it depends on an external service, but the application should show a loading indicator. 3D models should load without blocking the rest of the portfolio.

NFR3 – Security: User accounts and private portfolio information should be protected from unauthorized access. Authentication and authorization should be used for user-specific data. API keys and other credentials must not be exposed in the frontend or GitHub repository.

NFR4 – Privacy: Projectory should expose portfolio and career information according to the student's sharing choices. Information sent to the AI service should be limited to what is needed to generate career guidance.

NFR5 – Reliability and Fault Tolerance: The system should handle invalid input, database errors, unavailable AI services, and unsupported 3D models without crashing. Users should receive understandable error messages.

NFR6 – Availability: Core portfolio functions should remain available during normal expected usage. If an external AI or 3D-related service is temporarily unavailable, unaffected portfolio functions should remain usable.

NFR7 – Scalability: The application architecture should allow the system to support additional students, portfolios, AI requests, and 3D projects as usage increases. The initial class project is expected to support a small beta group, with the design allowing later growth.

NFR8 – Maintainability: Frontend and backend code should remain separated with clear APIs between them. Team development will use Git/GitHub branches and pull requests so changes can be reviewed before integration.

NFR9 – Compatibility: Projectory should function on current versions of Chrome, Firefox, Edge, and Safari and should provide a usable layout on common desktop and mobile screen sizes.

NFR10 – Accessibility: The interface should use readable text, labels, keyboard-accessible controls where practical, and reasonable contrast. Important project information should still be available in text when a 3D model cannot be viewed.

6. Competitive Analysis

Projectory overlaps with existing career and student-profile platforms, but the planned combination of a portfolio, portfolio-based AI guidance, and interactive 3D project presentation is the main area the team wants to explore.

Key Feature	Skills Council CareerGPS	Handshake	CareerGPSAI	Projectory
AI career guidance / recommendations	Yes	Yes	Yes	Yes
Skills or profile-based analysis	Yes	Yes	Yes	Yes
Student profile / portfolio	Limited	Yes	Yes	Yes
Project showcase	No clear focus	Yes	Limited	Yes
Job matching / opportunities	Yes	Yes	Job-focused	Not a core M1 feature
Interactive 3D project showcase	No	No	No	Yes

Projectory is planned specifically around a student's own portfolio rather than only job searching or general career exploration. Similar products already provide career recommendations, skills analysis, profiles, or job-related guidance. Projectory's planned advantage is connecting those ideas to a student project portfolio and allowing selected projects to include an interactive 3D model. This gives students one place to present what they have accomplished and then use the same portfolio information to receive AI-generated career paths, strengths, areas for improvement, and suggested next steps.

Competitor information reviewed from public product/help pages for Skills Council of Canada CareerGPS, Handshake, and CareerGPSAI (September 2026).

7. High-Level System Requirements

Component	Technology	Purpose
Frontend	React + Vite	Portfolio UI, project pages, career guidance views, and 3D viewer integration.
Backend	Python + FastAPI	Application APIs, business logic, database communication, and AI service requests.
Database	PostgreSQL hosted on Supabase	Store user, portfolio, education, skill, experience, and project data.
Frontend Deployment	Vercel	Host the React frontend.
Backend Deployment	Render	Host the FastAPI backend.
AI Technology	LLM API (provider TBD)	Analyze structured portfolio information and return career guidance. API calls handled through backend.
3D Technology	Niantic Scaniverse + browser-based 3D viewer (viewer framework TBD)	Create/use supported 3D project assets and display them interactively in the web application.
Version Control	Git + GitHub	Source control, branches, pull requests, reviews, and milestone documentation.
Supported Browsers	Chrome, Firefox, Edge, Safari	Primary modern browsers targeted by the web application.

The current M0 architecture will remain the base for M1. The React frontend will communicate with the FastAPI backend through APIs. FastAPI will communicate with PostgreSQL on Supabase and will handle requests to the selected LLM API so that credentials are not exposed in the frontend. The team will evaluate a browser-based 3D viewer such as Three.js or React Three Fiber before selecting the final viewer technology. Niantic Scaniverse will be investigated for producing supported 3D assets for project showcases.

8. Team

Team Member	Role	Study Area	One-Month Goal / Activity
Rohit Vennelakanti	Team Lead	Product direction / integration	Coordinate the M1-to-M2 scope, review integration between portfolio, AI, and 3D features, and help remove blockers as prototypes are developed.
Fanta Phommachith	Scrum Master	Team workflow / requirements	Maintain meeting notes and task tracking, help keep user stories

			current, and become familiar with the product flow so requirements can be updated as the team receives feedback.
Nuradil Kumarov	GitHub Master	Git/GitHub workflow	Maintain branch and pull-request workflow, milestone organization, and code-review practices. Help ensure each member can work from the shared repository without disrupting the stable branch.
Obaidullah Darwishi	AI Master	LLM integration	Prototype a structured portfolio-to-career-guidance request and response. Define the initial AI output for career paths, strengths, areas for improvement, and next steps.
Ali Coklu	Backend	FastAPI / Supabase	Implement initial portfolio and project CRUD flow and prepare backend endpoints needed by the frontend and AI feature.
Ian Ayala	Frontend	React / browser 3D	Build the first portfolio interface and test displaying one sample interactive 3D model in the React frontend using the selected browser-based viewer.

One-Month Technology Study Plan

Technology / Participants	Goal Within One Month
Frontend – Ian (lead), Fanta (support)	Build the initial portfolio page structure in React/Vite and test the flow for adding/editing portfolio information.
Backend/Database – Ali (lead), Nuradil (support)	Create the first FastAPI/Supabase data flow for core portfolio entities and confirm frontend-to-backend requests work in the deployed development environment.
Cloud/Deployment – Nuradil (lead), Rohit (support)	Keep the Vercel frontend and Render backend connected to the team repository and document the deployment/branch process for the team.
AI – Obaidullah (lead), Rohit (support)	Test one LLM API workflow using sample structured portfolio data and return a consistent response containing career paths, strengths, areas for improvement, and next steps.
3D/Niantic – Ian (lead), Obaidullah and Rohit (support)	Create or obtain one Scaniverse-generated sample asset and display it interactively in a browser proof of concept. Compare practical viewer options before the team commits to one.

9. Checklist

Current team status for the M1 checklist:

Checklist Item	Status
Team found a time slot to meet outside of class.	DONE
Scrum Master shares meeting minutes with everyone after each meeting.	ON TRACK
GitHub master chosen.	DONE
Everyone sets up their local development environment from the team's Git repository.	DONE
Team decided and agreed together on the listed software tools and deployment servers.	DONE
Team ready and able to use the chosen backend/frontend frameworks.	ON TRACK
Frontend study lead and one-month goal identified.	DONE
Backend/database study lead and one-month goal identified.	DONE
Cloud/deployment study lead and one-month goal identified.	DONE
AI study lead and one-month goal identified.	DONE
Niantic/3D study participants and one-month goal identified.	DONE
Team Lead ensured all team members read the final M1 and agree/understand it before submission.	ON TRACK